Uncertainty Estimation for Medical Image Registration

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1 Introduction

Over the years, Medical Image Registration (MIR) has been a vital tool and foundational component for all computer-assisted interventions and diagnostic radiology. The primary goal is to determine the optimal spatial transformation to register a source (moving) image I_m to a target (fixed) image I_f in a common coordinate space [Zitová and Flusser, 2003]. In clinical settings, this becomes a practical necessity rather than merely a theoretical task. It facilitates the integration of diverse imaging modalities and supports the precise, real-time tracking of surgical tools. For instance, in neuro-oncology surgery, a neurosurgeon must register preoperative high-contrast Magnetic Resonance Imaging (MRI) data with intra-operative Ultrasound (iUS) to correct for "brain shift" induced by craniotomy [Reinertsen et al., 2007]. This is particularly critical in high-stakes environments in which image registration is the lens through which a robotic system or neurosurgeon views the patient. Thus, if the lens is imperfect, a navigation system may attempt to steer a biopsy needle into healthy tissue rather than malignant tissue.

Realizing the importance of MIR, there has been a seismic shift in the field, transitioning from classical, iterative optimization-based approaches to learning-based ones. For decades, the field has employed techniques like Symmetric Normalization (SyN) [Avants et al., 2008] or B-spline Free Form Deformations (FFDs) [Rueckert et al., 1999]. While these approaches have been mathematically sound, they have been computationally expensive, making them unsuitable for real-time applications. The emergence of Deep Learning (DL) architectures, first by VoxelMorph [Balakrishnan et al., 2019] and later refined by foundation models like UniGradICON [Tian et al., 2024], has changed the landscape by shifting the heavy lifting to a training phase. This transition has successfully reduced computation times from hours to milliseconds, making complex deformable registrations suitable for live surgical feedback.

Nonetheless, these speed improvements have been at a substantial cost to transparency and reliability. Classical techniques typically offer a convergence energy or a cost function value that provides some insight into the quality of the registration. In contrast, most deep learning registration networks are black-box systems that behave as deterministic point estimators. They offer a "best guess" at a displacement field without offering any insight into their own unreliability. In safety-critical clinical environments, these networks often suffer from overconfidence and offer no hindication that the registration has failed due to occlusions, low signal-to-noise ratios, or unusual patient anatomy. This absence of a reliability metric is the main limiting factor for the adoption of DL registration in the operating-room environment.

Addressing these transparency issues calls for designing registration systems that are not only accurate but also self-aware of potential inaccuracies. Indeed, a truly robust system should have the capability to alert the clinician when predictions are likely to be in error, particularly in regions near critical structures. To close this gap, this investigation aims to integrate robust Uncertainty Quantification (UQ) techniques into deep learning frameworks [Abdar et al., 2021]. By moving beyond deterministic

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outputs and providing a probabilistic understanding of the displacement field, we can ensure that medical image registration transitions from a black-box predictive tool to a reliable diagnostic one.

2 Background and Problem Formulation

2.1 Deformable Registration

Let I_f be the fixed (target) image and I_m be the moving (source) image defined over a d-dimensional domain $\Omega \subset \mathbb{R}^d$. Deformable registration is formulated as a transformation $\phi = \mathrm{Id} + u$, where Id is the identity mapping and u is a displacement field, such that:

$$I_f \approx I_m \circ \phi \tag{1}$$

In a learning-based framework, a neural network f_θ is trained to predict ϕ by minimizing a loss function $\mathcal{L}$:

$$\mathcal{L}(I_f, I_m, \phi) = \mathcal{L}_{\text{sim}}(I_f, I_m \circ \phi) + \lambda R(\phi) \tag{2}$$

Similarity loss ($\mathcal{L}_{\text{sim}}$), such as Normalized Cross-Correlation (NCC) or Mean Squared Error (MSE), assesses the anatomical similarity between images, while the regularization term (R), often based on the gradient of the displacement field, enforces the registration solution to satisfy physical feasibility constraints such as non-folding/non-tearing of the tissue (maintaining a positive Jacobian determinant, $|J_\phi| > 0$).

2.2 The Paradox of Ground Truth

The main challenge in evaluating MIR is the inherent absence of a Ground Truth (GT) for the displacement field, referred to as ϕ_{true}. Unlike an image classification task, where a human can definitively label a cat, there is no way for an expert radiologist to definitively determine the voxel-wise correspondence between two dense 3D images of the brain. This gives rise to the Correspondence Paradox:

- To calculate the true Registration Error ($E = \|\phi_{true} - \phi_{pred}\|$), we must know ϕ_{true}.
- Conversely, if ϕ_{true} were known with mathematical certainty, the registration task would be redundant.

Furthermore, existing clinical metrics, such as Dice Scores or Landmark Distances, provide limited information on accuracy [Crum et al., 2006, Fitzpatrick et al., 1998]. Dice scores measure the overlap of entire anatomical structures but ignore internal voxel-wise misalignment, whereas landmark distances provide high-accuracy data but only at a few discrete points. Consequently, because these clinical proxies fail to capture the dense, voxel-wise reality of the deformation, we are left in a position where the actual Registration Error remains mathematically hidden during inference. This fundamental lack of dense supervision necessitates a conceptual and practical distinction between Registration Error (the unobservable distance to ϕ_{true}) and Predictive Uncertainty (statistical variance in ϕ_{pred}), which is further subclassified into the components Aleatoric (data-inherent noise) and Epistemic (model-driven inconsistency) [Kendall and Gal, 2017].

To address this limitation, we introduce the concept of a Synthetic Oracle bridge. By using the `torchio` [Pérez-García et al., 2021] library to apply known transformations to 3D MRI volumes, we create a supervised setting. This allows us to move from the problem of estimating Uncertainty to the problem of Error Map Regression.

3 Related Works

The existing research on Uncertainty Quantification (UQ) for Medical Image Registration (MIR) can be generally classified into three methodological pillars: sampling-based approximations, probabilistic architectures, and consistency-based self-supervision. However, as established by the Correspondence

Paradox (Section 2.2), the lack of ground truth makes it difficult to verify if the uncertainty signals provided by these methods are truly representative of model reliability.

3.1 Sampling-Based Uncertainty (Dropout and Ensembles)

The most prevalent approach to UQ in registration is the use of Monte Carlo (MC) Dropout, popularized by the Quicksilver framework [Yang et al., 2017]. This method uses the Bayesian interpretation of dropout as a Gaussian process approximation to obtain a distribution of displacement fields ϕ by passing the network forward several times during the inference phase [Gal and Ghahramani, 2016]. The variance of this distribution is used to represent the uncertainty.

The fundamental critique of sampling-based methods in the UQ domain is the assumption that variance equals unreliability. While variance is successful in pointing to areas where the model is confused or is sensitive to changes in the weights, it does not identify areas where the model is systematically biased. The model may have high confidence (low variance) in the prediction, but it may be systematically incorrect. Without an objective reference point to anchor the variance maps, it is not clear if a particular region is actually a correct region.

3.2 Probabilistic and Latent Variable Models

Probabilistic U-Nets [Kohl et al., 2018] and similar cVAE-based structures attempt to model Aleatoric uncertainty (data-inherent ambiguity). By modeling the distribution $P(z \mid I_f, I_m)$, they can sample different possible anatomical alignments. This is theoretically sound to model the one-to-many mapping issue in ambiguous regions with weak boundaries or low texture.

However, from a UQ perspective, these probabilistic frameworks face a major calibration challenge. While a probabilistic model can provide a probability distribution, it does not ensure that this probability distribution is well-calibrated to the actual anatomical truth. If the learned prior is wrong, it can provide a tight (high-confidence) distribution around an incorrect warp. In this case, we are using uncertainty as a measure of internal plausibility rather than clinical reliability.

3.3 Consistency-Based UQ: Transformation Equivariance

A more recent paradigm utilizes Transformation Equivariance [Tian et al., 2025] as a self-supervised uncertainty signal. This method measures Model Self-Consistency, which means if the input images are transformed (e.g., rotation), then the resulting warp should follow that same transformation. The Equivariance Error serves as a proxy for uncertainty, eliminating the need for multiple forward passes.

A limitation of this approach is the distinction between consistency and accuracy. While equivariance is a necessary condition for a robust model, it is not a sufficient condition. A registration model could be perfectly equivariant (highly consistent) but still produce a displacement field that is anatomically incorrect across all rotations. Thus, while equivariance is a first-rate measure of detecting instabilities, its validity as a measure of uncertainty is limited by its inability to detect stationary biases in the foundation model's architecture.

3.4 Identifying the Gap: Grounding Consistency in Correctness

The inherent limitation of current UQ methods, from sampling-based variance to geometric equivariance, is that they mainly quantify the model's inconsistency (precision) without considering the anatomical correctness (accuracy). This is because these frameworks are based on the Correspondence Paradox and cannot verify whether their model's confidence is actually related to the Registration Error (E). This is a critical clinical issue because a model may have high confidence (low variance/inconsistency) yet be fundamentally wrong. Such a model will make certain yet wrong predictions that cannot be captured during inference.

In this research study, we are exploring this gap in the current understanding of UQ frameworks through the use of a synthetic Oracle Bridge to verify whether the confidence of a model is related to the registration error. By using the registration error map of a foundation model such as UniGradICON [Tian et al., 2024] as a learnable diagnostic feature, we are exploring whether the inherent structural characteristics of registration failure can be captured using a secondary neural network. This represents a systematic verification of whether a model's self-reported confidence accurately reflects its true anatomical accuracy.

4 Methodology: The Three-Stage Workflow

To evaluate the relationship between registration error and predictive uncertainty, we built a modular pipeline designed to bypass the Correspondence Paradox. Rather than attempting to estimate uncertainty directly from unobservable ground-truth correspondences in clinical acquisitions, we apply controlled synthetic spatial transformations on real 3D anatomical structures to construct precise, dense ground-truth displacement fields u_{gt}. We then formulate registration error map prediction as a supervised dense regression task.

As illustrated in Figure 1, the proposed pipeline comprises three sequential phases:

1. **Phase 1: Ground Truth Data Generation:** Synthetic dense transformation fields are synthesized and applied to 3D anatomical volumes.
2. **Phase 2: Registration Error Map Extraction:** Displacement field differences are evaluated on the native source lattice.
3. **Phase 3: Supervised Error Map Training:** A deep 3D convolutional neural network (U-Net) is trained to regress the spatial error distribution $\hat{y}$.

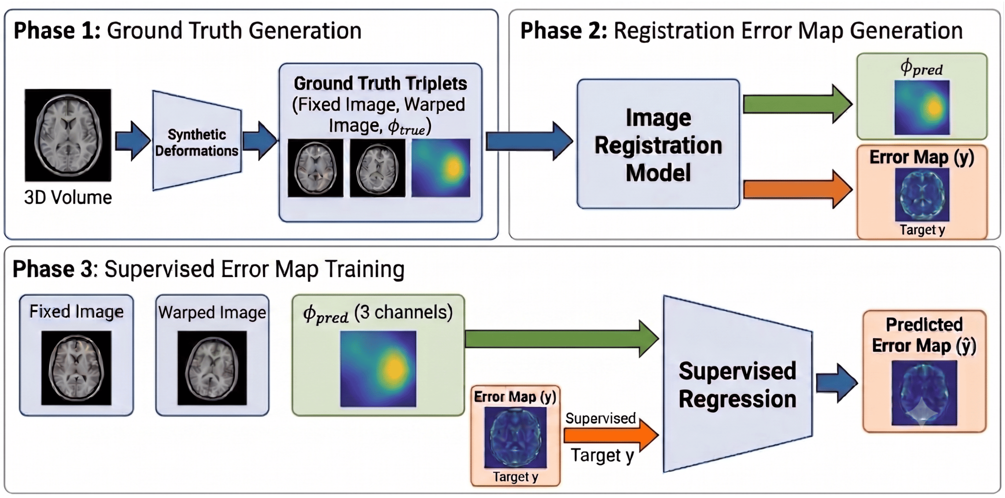

Figure 1: Overview of the three-stage methodology pipeline for registration error map prediction. The workflow includes ground-truth generation via synthetic transformations (Phase 1), extraction of registration error maps (Phase 2), and supervised regression training to produce predicted error maps $\hat{y}$ (Phase 3).

4.1 Phase 1: Ground Truth Generation

Phase 1 constructs paired 3D brain volumes with exact, voxel-wise ground-truth displacement vectors $u_{\text{gt}} \in \mathbb{R}^{3\times H\times W\times D}$.

We use minimally preprocessed T1-weighted (T1w) brain MRI scans from the Human Connectome Project (HCP) Young Adult 2025 dataset [Van Essen et al., 2013, WU-Minn HCP Consortium, 2025]. Specifically, we take the skull-stripped volumes from each subject's `T1w/` directory (not the scanner-native *unprocessed/* data). These volumes have already undergone HCP's standard minimal preprocessing: AC-PC alignment, gradient and readout (spatial) distortion correction, and bias-field correction, while remaining in native subject space at approximately 0.7 mm isotropic resolution. Keeping native space preserves inter-subject anatomical variability; nonlinear normalization into a template, such as MNI152, would reduce that variability and make medical image registration a less realistic test for our setup.

From the subjects that have complete T1w scans and valid FreeSurfer brain masks, we select 1113 volumes. We split the dataset deterministically into training, validation, and test sets using a $75/10/15$ split policy with a fixed seed (seed 42). The resulting counts are shown in Table 1. Representative axial views of minimally preprocessed HCP T1w anatomy are shown in Figure 2.

Table 1: Partition statistics for the HCP Young Adult cohort utilized in Phase 1 ground-truth generation.

Dataset Split	**3D Volumes (N)**	**Proportion (%)**
Train	857	77.0
Validation	109	9.8
Test	147	13.2
Total	1113	100.0

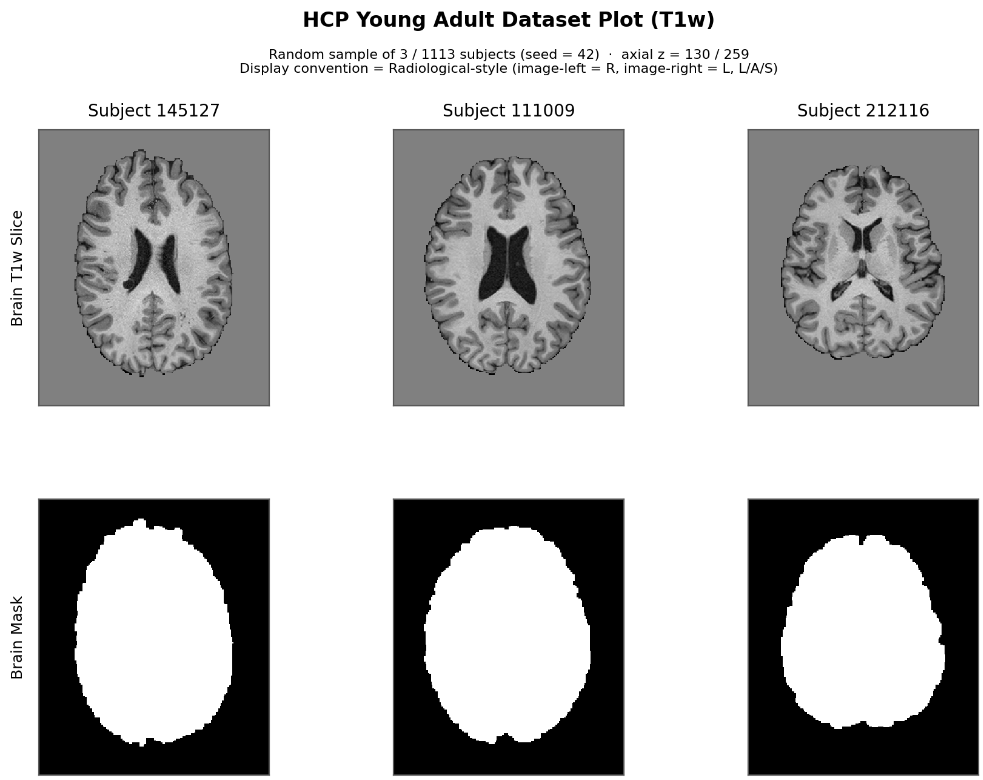

Figure 2: Representative HCP T1w volumes (skull-stripped) used as fixed anatomy in Phase 1.

4.1.1 TorchIO-Based Synthetic Transformations and Ground-Truth Displacements

For each subject, we synthesize a single registration pair where the native T1w volume serves as the fixed target image I_{source}. Synthetic transformations are generated in physical coordinate space (mm) using TorchIO. By leveraging the NIfTI spatial affine, TorchIO applies controlled spatial transformations to construct the warped moving volume I_{moving}. We store a corresponding dense ground-truth displacement field u_{gt} on the fixed source lattice in voxel units, satisfying:

$$I_{\text{moving}}(x + u_{\text{gt}}(x)) \approx I_{\text{source}}(x) \tag{3}$$

Operationally, sampling on the identity grid produces a temporary backward field. We invert this field using SimpleITK to obtain the forward displacement map u_{gt}. Intensity volumes are normalized using masked z-score normalization, leaving spatial coordinates and displacement vectors unaltered. Following field inversion, we clean the ground-truth displacement maps u_{gt} in three steps:

1. Zeroing out invalid voxels outside the identity grid mask.
2. Applying a zero-value 12-voxel border along each volume face.
3. Clipping total displacement magnitudes $\|u_{\text{gt}}\|$ at the 99.9th percentile over all non-zero voxels to remove extreme spatial outliers.

Every subject is assigned to one of five transformation classes. We set target split ratios to 5% identity (`none`), 20% `rigid`, 25% `affine`, 25% `elastic`, and 25% composite `affine_elastic`. Within each dataset partition, subjects are shuffled using a split-specific seed, and exact class quotas are calculated using largest-remainder rounding. The realized subject distributions across splits are summarized in Table 2.

Table 2: Realized transformation-class counts per split (one sample per HCP subject).

Class	Train	Val	Test	Total	Target Share
none	43	6	7	56	5%
rigid	172	22	29	223	20%
affine	214	27	37	278	25%
elastic	214	27	37	278	25%
affine + elastic	214	27	37	278	25%
Total	857	109	147	1113	100%

Table 3 details the parameter ranges for each active class. The `rigid` regime applies uniform rotations ($\leq 8.0°$) and translations (≤ 5.0 mm) while holding scaling factors at unity. The `affine` regime incorporates isotropic scale sampling (0.96 to 1.04), expanded global rotations ($\leq 10.0°$), and translations (≤ 5.5 mm). Non-linear deformations in the `elastic` class use B-spline fields with seven control points per axis, a maximum displacement of 12.0 mm, and two locked border layers to prevent boundary artifacts. The `affine_elastic` class composes the baseline affine parameters with a milder elastic deformation (maximum displacement of 8.0 mm).

Table 3: Transformation parameter envelopes used for this HCP synth cohort (physical units; TorchIO samples uniformly within each range). The `none` class applies no transform.

Class	TorchIO Sampling Parameters
none	no transformation
rigid	scales fixed at 1.0; degrees $\pm 8°$; translation ± 5.0 mm (each axis)
affine	scales $\in [0.96, 1.04]$; degrees $\pm 10°$; translation ± 5.5 mm (each axis)
elastic	max displacement 12.0 mm (each axis); 7 control points; locked borders $= 2$
affine + elastic	affine as above; then elastic with max displacement 8.0 mm

4.1.2 Ground Truth Dataset Statistics

To assess the distribution and visual fidelity of the synthesized ground-truth displacement fields, we evaluate per-class $\|u_{\text{gt}}\|$ metrics across the dataset and plot representative qualitative samples. Table 4 summarizes the per-class $\|u_{\text{gt}}\|$ statistics computed over the full cleaned volume (excluding the 12-voxel zeroed border) on the Train split. Identity (`none`) samples contribute zero displacement by construction, whereas the `rigid`, `affine`, and `affine_elastic` regimes induce larger mean displacement magnitudes than pure `elastic` transformations under our parameter envelopes. Similar ground truth displacement statistics are observed across the validation and test partitions, reflecting stable sampling across all splits.

Figure 3 displays orthogonal quality control (QC) views for two representative synthetic training cases (`elastic` and `affine_elastic`), showing the source image, displacement overlay, moving image, ground-truth displacement magnitude $\|u_{\text{gt}}\|$, and checkerboard alignment maps. To illustrate the range of transformation severity, Figure 4 ranks non-identity training samples by mean $\|u_{\text{gt}}\|$, contrasting easy, typical, and hard cases across different transformation classes.

Table 4: Phase 1 Train split: per-class $\|u_{\mathrm{gt}}\|$ summary statistics (voxels).

Class	n	**Q1**	**mean**	**Q3**	**max**
none	43	0.00	0.00	0.00	0.00
rigid	172	2.51	10.76	16.90	29.42
affine	214	0.82	12.95	20.58	36.85
elastic	214	0.82	3.13	4.90	9.98
affine + elastic	214	1.01	13.27	21.03	37.35

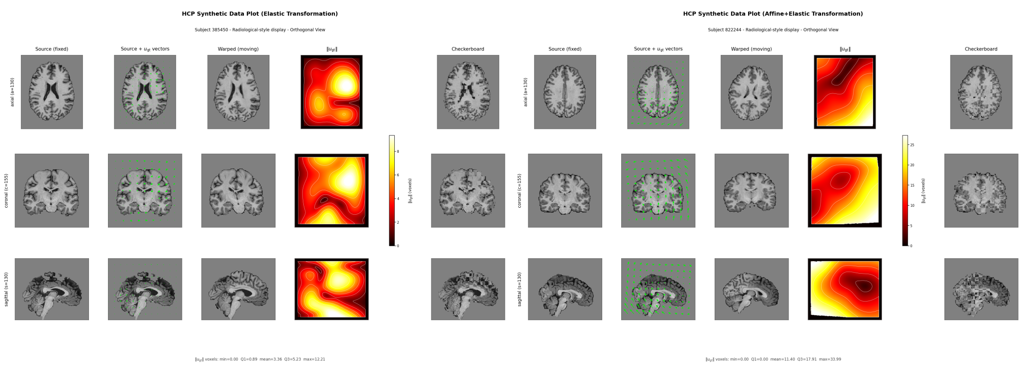

Figure 3: **Representative HCP synthetic triplet (Train elastic, and affine + elastic).** Orthogonal mid-slices of the fixed source, displacement overlay, warped moving image, ground-truth displacement magnitude $\|u_{\mathrm{gt}}\|$ in voxels, and checkerboard map.

4.2 Phase 2: Registration Error Map Generation

Phase 2 constructs a voxel-wise supervised target by measuring the spatial discrepancy between the foundation model, UniGradICON [Tian et al., 2024], and the ground-truth displacement map u_{gt}. For every synthesized sample from Phase 1, we execute UniGradICON as $\texttt{net}(I_{\mathrm{moving}}, I_{\mathrm{source}})$ to obtain a predicted displacement field u_{pred} defined on the native source lattice. Following the prediction, we apply identical border zeroing and percentile clipping procedures to u_{pred} as performed in Phase 1 to maintain coordinate consistency. We define the dense registration error map y in voxel units as the Euclidean distance:

$$y(x) = \|u_{\mathrm{gt}}(x) - u_{\mathrm{pred}}(x)\| \tag{4}$$

Both the anatomical MRI volumes and the UniGradICON foundation model operate natively in 3D physical space. Consequently, u_{pred} is evaluated and stored directly on the 3D source grid, matching the exact spatial dimensions and coordinate orientation of u_{gt}. Processing volumes natively in 3D avoids spatial resampling artifacts and preserves dense voxel alignment between the ground truth and predicted displacement fields.

Figure 5 illustrates representative Phase 2 outputs for two synthetic training cases (`elastic` and `affine + elastic`), detailing the source and moving anatomy, ground-truth magnitude $\|u_{\mathrm{gt}}\|$, predicted magnitude $\|u_{\mathrm{pred}}\|$, the registration error map y, and the directional cosine similarity $\cos\theta$. The directional alignment map confirms that u_{pred} and u_{gt} point in consistent spatial directions across most anatomical regions, showing values near $+1$, which demonstrates that y primarily measures local magnitude discrepancy. To evaluate the variation in registration difficulty across the cohort, Figure 6 ranks training volumes by mean y, contrasting easy, typical, and hard cases across different transformation classes. Phase 2 thus provides the dense ground-truth regression target y, leaving only observable inputs, namely the anatomical image pair and u_{pred}, as candidate features for Phase 3 model training.

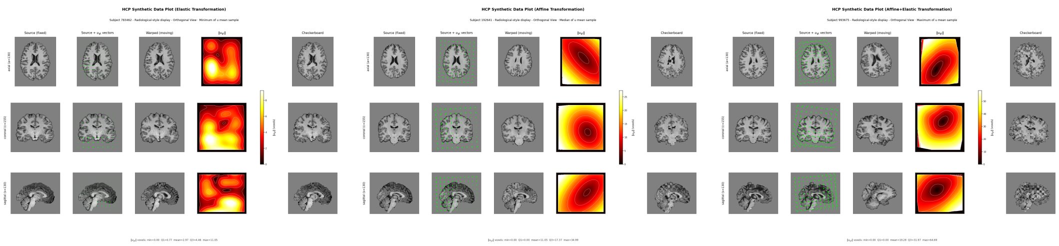

Figure 4: **Synthetic triplets by transformation difficulty (Train).** Left to right: minimum, median, and maximum mean $\|u_{\text{gt}}\|$, contrasting easy, typical, hard cases among non-identity samples.

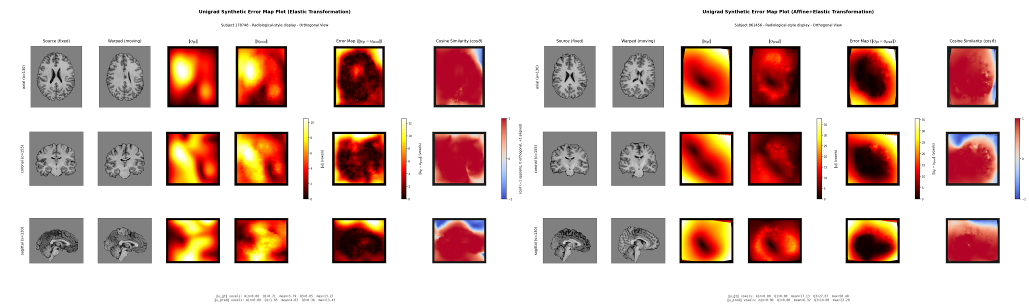

Figure 5: **Synthetic ground truth versus UniGradICON predictions (Train: elastic, and affine + elastic).** Orthogonal views of source and moving anatomical volumes alongside displacement magnitudes $\|u_{\text{gt}}\|$, $\|u_{\text{pred}}\|$, the dense registration error map $y = \|u_{\text{gt}} - u_{\text{pred}}\|$ in voxels, and voxel-wise cosine similarity $\cos\theta$ indicating vector alignment.

4.3 Phase 3: Supervised Error Map Training

Given the target error maps y generated in Phase 2, Phase 3 trains a 3D U-Net to regress the scalar error map strictly from observable inputs available during inference: I_{source}, I_{moving}, and u_{pred}.

4.3.1 Data Setup and Preprocessing

The data set partitions established in Phase 1 are strictly maintained: the training and validation splits are used for optimization and hyperparameter tuning, while the test split ($n = 147$) remains unchanged. Each input volume provides five spatial channels $[I_{\text{source}}, I_{\text{moving}}, u_{\text{pred},x}/s, u_{\text{pred},y}/s, u_{\text{pred},z}/s]$ with a fixed scaling factor $s = 64$ voxels, accompanied by the target registration error map $y = \|u_{\text{gt}} - u_{\text{pred}}\|$ and a binary brain mask `source_mask` restricting loss computation and evaluation metrics to brain tissue. Volume intensities retain their masked z-score normalization from Phase 1.

4.3.2 U-Net Architecture

We implement a 3D encoder-decoder U-Net architecture equipped with skip connections and Group Normalization. The network receives the padded five-channel volume at the input, and emits a single-channel prediction $\hat{y}$ at full spatial resolution, utilizing internal spatial padding to the nearest multiple of 16 before the encoder and central cropping at the output, returning to the native lattice. The encoder consists of four successive pooling stages. Feature channels scale relative to a base channel width $b = 16$, progressing through b, $2b$, $4b$, and $8b$ channels, culminating in a $16b$ bottleneck. The decoder mirrors this hierarchy using transposed convolutions, where a final $1 \times 1 \times 1$ convolution yields the predicted error map $\hat{y}$. The volumetric 3D design natively operates on the spatial grid of the HCP anatomy and the UniGradICON displacement fields, preserving 3D spatial continuity without requiring slice-wise decomposition.

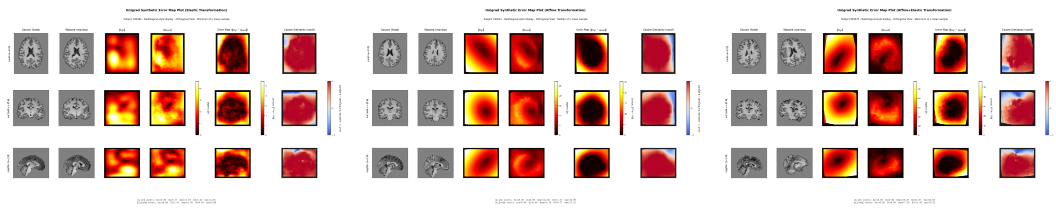

Figure 6: **Registration error maps by difficulty (Train).** Left to right: minimum, median, and maximum mean registration error y, contrasting easy, typical, and hard cases among non-identity samples.

4.3.3 Training Strategy

The training objective is defined as the masked mean absolute error (MAE) over the set of valid brain voxels Ω_{valid} designated by `source_mask`:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\Omega_{\text{valid}}|} \sum_{x \in \Omega_{\text{valid}}} \left| \hat{y}(x) - y(x) \right| \tag{5}$$

Optimization is performed using AdamW with an initial learning rate of 10^{-3} and weight decay of 10^{-5}, accelerated via CUDA mixed precision training. Learning rates are adjusted using a ReduceLROnPlateau scheduler based on validation performance, and early stopping is enforced on validation MAE. Validation is conducted every three epochs beginning at epoch three, with the lowest validation MAE checkpoint retained for final testing. Training and validation loss trajectories across an 80 epoch budget (early-stopped) are illustrated in Figure 7.

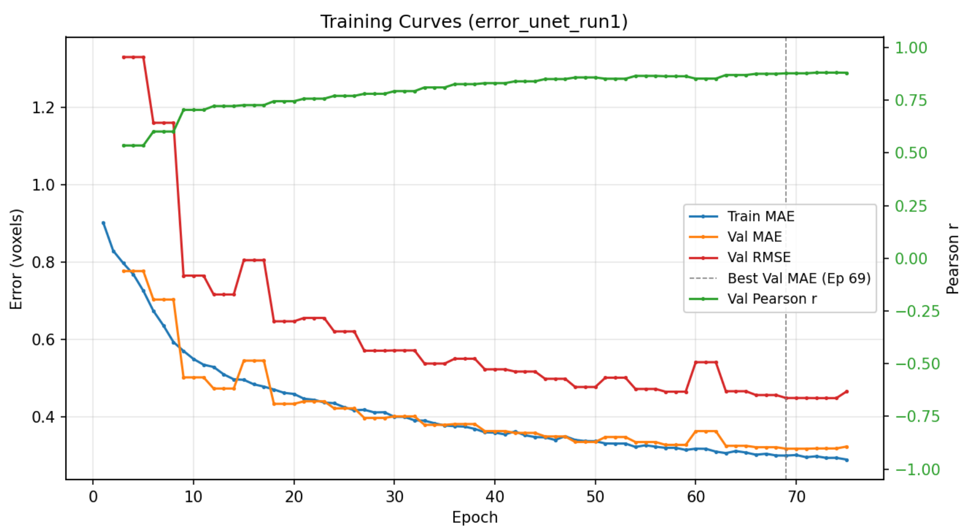

Figure 7: **Phase 3 training curves (HCP 3D U-Net).** Masked training and validation MAE over epochs. The selected model checkpoint corresponds to the global minimum validation MAE.

4.3.4 Quantitative and Qualitative Evaluation

The optimal checkpoint is evaluated on the held-out test set ($n = 147$) using `source_mask`. Aggregate masked metrics are summarized in Table 5: a masked MAE of 0.338 voxels, a masked RMSE of 0.485 voxels, and a Pearson correlation coefficient of $r = 0.874$ between predicted $\hat{y}$ and true error y. These quantitative results demonstrate that dense UniGradICON registration error distributions are predictable from observable image pairs and predicted displacement fields alone.

Qualitative test evaluation is presented across two visual perspectives: (i) two representative transformation classes (`elastic` and `affine + elastic`; Figure 8), and (ii) minimum, median, and maximum test cases ranked by mean masked target error (Figure 9).

Table 5: Phase 3 Test performance on held out HCP cohort ($n = 147$, evaluated within `source_mask`, units in voxels).

Split	Masked MAE	Masked RMSE	Pearson r
Test ($n = 147$)	0.338	0.485	0.874

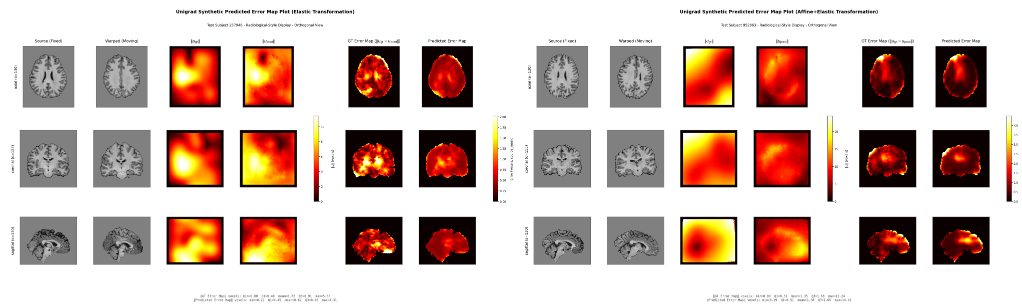

Figure 8: **Qualitative Test evaluation (elastic, and affine + elastic samples).** Orthogonal views of source and moving anatomy, $\|u_{\text{gt}}\|$, $\|u_{\text{pred}}\|$, ground-truth error map y, and U-Net error prediction $\hat{y}$ (in voxels).

4.4 Summary of Workflow Artifacts

Table 6 provides an overview of the three-stage pipeline, summarizing the primary input and output artifacts generated at each methodology phase.

Table 6: Summary of the three-stage methodology pipeline and primary supervision artifacts.

Phase	Primary Pipeline Output and Artifacts
Phase 1: Ground Truth Generation	Synthetic HCP volumes containing $(I_{\text{source}}, I_{\text{moving}}, u_{\text{gt}})$, brain masks, and transformation class labels. Displacements are stored in voxel units on the native lattice.
Phase 2: Error Map Generation	UniGradICON predicted fields u_{pred} and corresponding dense registration error maps $y = \|u_{\text{gt}} - u_{\text{pred}}\|$ evaluated on the source grid.
Phase 3: Supervised Training	A 3D U-Net regressor trained with masked MAE to predict $\hat{y}$ from observable features $[I_{\text{source}}, I_{\text{moving}}, u_{\text{pred}}/s]$. The optimal checkpoint is selected via validation MAE.

The complete source code implementing this three-stage framework, including synthetic volume generation, foundation model evaluation, and 3D U-Net regression, is publicly accessible on GitHub[1].

5 Analysis

This empirical evaluation addresses a core question underlying the Correspondence Paradox: when dense registration error is rendered observable via a synthetic oracle, can that registration error map be recovered from signals available at inference time, specifically the source image, moving image, and predicted displacement field? The Phase 3 results on the benchmark dataset provide an affirmative answer within this controlled experimental framework.

Quantitative Behavior: On the held-out test split ($n = 147$), the 3D U-Net achieves a masked MAE of 0.338 voxels, a masked RMSE of 0.485 voxels, and a Pearson correlation coefficient of $r = 0.874$ relative to target y within `source_mask` (Table 5). The strong linear correlation is particularly informative, demonstrating that the network learns the underlying spatial structure of foundation

[1] `https://github.com/kubershahi/uncertainty-quantification`

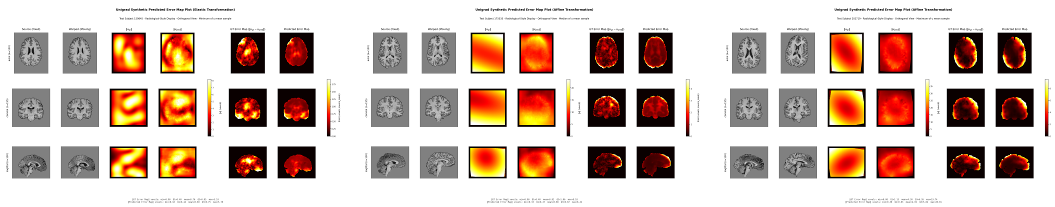

Figure 9: **Phase 3 Test evaluation across error difficulty.** Left to right: minimum, median, and maximum masked mean target error y, contrasting easy, typical, and hard cases among non-identity samples.

model registration errors rather than merely regressing toward a global spatial mean. Training dynamics align with this interpretation (Figure 7): training MAE decreases from approximately 0.90 to 0.29 voxels, while validation MAE, RMSE, and Pearson r improve monotonically across evaluation intervals. The optimal validation MAE occurs at epoch 69 ($\text{val MAE} \approx 0.318$, $\text{val}\, r \approx 0.877$), closely matching final test performance and indicating strong generalization under the subject-level split.

Qualitative Error Localization: Orthogonal visualizations on held-out test volumes (Figures 8 and 9) confirm that predicted error fields $\hat{y}$ reliably track local maxima in ground-truth registration fields y. Registration errors are spatially localized rather than uniform, and the U-Net concentrates response magnitude precisely where the registration model deviates from ground truth u_{gt}. Low error cases yield near-zero predictions, whereas high registration cases preserve distinct localized peaks. This behavior is ideal for diagnostic error monitoring, providing localized spatial alerts rather than diffuse, uninformative uncertainty distributions.

Implications for Uncertainty Quantification As discussed in Section 3.4, sampling variance and equivariance metrics primarily evaluate model self-consistency rather than true anatomical accuracy. The proposed methodology does not claim to serve as a standalone probabilistic uncertainty quantification framework. Instead, it demonstrates that once an oracle registration error $y = \|u_{\text{gt}} - u_{\text{pred}}\|$ is generated, that error becomes a learnable function of inference time observables. Failures in spatial registration leave systematic signatures in the input tuple $(I_{\text{source}}, I_{\text{moving}}, u_{\text{pred}})$. Establishing this predictability is a necessary prerequisite before calibrating self-reported uncertainty estimates against physical registration error.

Limitations: Several factors bound the scope of these empirical results. First, ground truth fields u_{gt} are generated via synthetic spatial transformations (rigid, affine, elastic, and composite formulations) rather than true inter-subject anatomical correspondence; performance on synthetic registration errors does not guarantee direct transfer to clinical registration scenarios. Second, predicting target y via supervised loss constitutes deterministic error regression rather than calibrated probabilistic uncertainty quantification (neither aleatoric nor epistemic). Third, the reported training setup (five channels, u-scale 64, GroupNorm, b = 16, masked MAE) employs a fixed hyperparameter configuration without ablations that isolate individual channel contributions or class-specific performance variations. Fourth, evaluations are restricted to a single structural imaging modality and a specific foundation model checkpoint, leaving cross-modality and cross-domain generalization unexamined. Finally, interpreting voxel-level MAE requires accounting for the spatial residual scale. Global averages are heavily influenced by well-aligned tissue where ground-truth error is near zero; thus, a masked MAE of ~ 0.34 voxels reflects high estimation fidelity in isolating localized registration failures, as confirmed by the strong spatial correlation.

Future Directions: These findings support the use of dense error maps as diagnostic supervision targets when oracle signals are available, as well as serving as baseline reference maps when auditing heuristic uncertainty estimators. Future research directions include per-class and per-region error breakdowns, channel ablation studies, direct comparison of predicted error maps against Monte Carlo dropout or equivariance maps, and extension to real anatomical pairs utilizing proxy supervision.

6 Conclusion

We presented a three-stage methodology for evaluating registration accuracy in settings where dense ground-truth is typically inaccessible. By coupling synthetic spatial transformations with deep learning registration networks, registration error maps $y = \|u_{\text{gt}} - u_{\text{pred}}\|$ serve as targets for a 3D U-Net trained strictly on inference time observables. Evaluated on 147 held out test volumes, the model achieves a masked Pearson correlation of $r = 0.874$ and a masked MAE of 0.338 voxels without severe overfitting. These results establish that registration errors produced by foundation models under synthetic warps are spatially structured and highly predictable. This provides a concrete foundation for anchoring future uncertainty quantification models directly in registration correctness rather than model consistency alone. Closing the loop to clinical uncertainty quantification will require moving beyond synthetic oracles, calibrating probabilistic uncertainty against such error maps, and validating under real anatomical and multimodal shifts.

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